

Skills Summary

Where Two Lines Meet

Use this reference while completing your worksheet or when studying.

Skill 1 — Testing a pair in both equations

Rule: a pair solves a system only if it makes *every* equation true. Satisfying one equation and failing the other means the point is on one line but not the other, so it is not the meeting point.

Example: Solve $x + y = 7$ and $x - y = 1$.

Step 1. Both equations must hold at once, so combine them rather than solving either one on its own — adding them cancels y .

Step 2. $2x = 8$, so $x = 4$; then $4 + y = 7$ gives $y = 3$. Check the second: $4 - 3 = 1$. Both true, so this is the solution. **(4, 3)**

Try it: Solve $x + y = 9$ and $x - y = 3$.

(8, 9) :check:

Skill 2 — Reading the intersection off a graph

Rule: the crossing point lies on both lines, so its coordinates satisfy both equations. Read the point, then substitute to confirm it.

Example: Find the solution of $y = x + 2$ and $y = 6 - x$ from the graph.



Step 1. The lines cross at a grid corner, so read the coordinates there and then confirm them in both equations.

Step 2. The corner is 2 across and 4 up: $4 = 2 + 2$ and $4 = 6 - 2$, both true. On both lines at once. **(2, 4)**

Try it: Solve $y = x + 3$ and $y = 9 - x$.

(9, 3) :check:

Skill 3 — Finding the shared row in tables

Rule: when two tables share an input with the same output, that row is the solution. If no row is shared, set the two rules equal and solve.

Example: Plan A is in the table; plan B follows $y = h + 12$. When do they cost the same?

h	1	2	3
Plan A	9	12	15

Step 1. Plan A's table rises 3 for each hour and starts from 6 at zero hours, so its rule is $y = 3h + 6$ and the two rules can be set equal.

Step 2. $3h + 6 = h + 12$ gives $2h = 6$, so $h = 3$; the table's cost there is **15**.

The shared row is the solution. **(3, 15)**

Try it: A table gives $h = 1, 2, 3$ with costs 8, 13, 18 (rule $y = 5h + 3$); the other plan follows $y = h + 15$. Find the shared pair.

(81, 18) :check

Skill 4 — What the intersection means in a story

Rule: the intersection is the moment the two options are equal. On one side of it the first option is cheaper; on the other side the second one is.

Example: Club A charges \$10 to join plus \$4 a visit ($y = 4v + 10$); club B charges \$6 a visit ($y = 6v$). When are they equal?

Step 1. Equal cost means the two expressions take the same value, which is the intersection of their graphs — set them equal.

Step 2. $4v + 10 = 6v$ gives $2v = 10$, so $v = 5$, and the cost there is $6(5) = 30$ dollars.

Below 5 visits club B is cheaper; above it, club A. **(5, 30)**

Try it: One club charges $y = 3v + 12$ and another charges $y = 5v$. Find the (visits, cost) pair where they are equal.

(03, 9) :check